223 results for “survey review wildfire ”forest fire” “machine learning” ”artificial intelligence” ”deep learning” techniques management”

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A Deep Learning Based Experiment on Forest Wildfire Detection in Machine Vision Framework

Spoorthi R · R. Uma · Nirmitha H · A. R · Environmental Science, Computer Science ·

International Journal of Scientific Research in… · 24 December 2025

TLDR This project demonstrates how **machine** vision and **artificial intelligence** technologies can be used to convert conventional **wildfire** monitoring systems into intelligent, data-driven solutions. Expand

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Innovations in Wildfire and Smoke Detection: A Comprehensive Survey

Deepa Mariam Thomas3 · Environmental Science, Computer Science ·

INTERANTIONAL JOURNAL OF SCIENTIFIC RESEARCH IN… · 28 February 2025

TLDR Progress in **fire** and smoke detection is explored, emphasizing the role of **machine learning** and computer vision **techniques** such as Convolutional Neural Networks (CNNs) and YOLO object detection models, and promising research directions for developing robust and scalable **wildfire** detection systems are identified. Expand

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Predicting Student Academic Performance with Machine Learning: A Systematic Literature Review

M. Chek · I. L. Ismail · Jamal. N · Z. Zulkifli · R. Nariswari · M. S. Asrulsani

Computer Science, Education · International journal of research and innovation… · 2026

TLDR R of **machine learning** approaches used for predicting student performance in higher education indicates that Random **Forest**, Support Vector **Machines** (SVM), Decision Trees, Logistic Regression, and **Artificial** Neural Networks are the most frequently used algorithms for student performance prediction. Expand

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Performance Monitoring of Photovoltaic Modules Using Machine-Learning-Based Solutions: A Survey of Current Trends

N. Nyangiwe · A. Kenfack · N. Thantsha · Mandla Msimanga

Engineering, Environmental Science · Energy Science & Engineering · 22 December 2025

TLDR Besides **review**ing new advancements on ML-based solutions, this paper also looks at key challenges facing the field, such as quality and availability of data, computation costs and robust interpretable ML models, and suggests strategies for addressing these challenges. Expand

2 · Wiley · Save · Cite

Early Natural Disaster Prediction Using Machine Learning: A Comprehensive Review

H. Yadav · Environmental Science, Computer Science ·

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN… · 30 April 2025

TLDR The capacity of these models across various disaster types, their dependency on both essential data sources, and their performance in overcoming existing and future challenges are assessed. Expand

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Optimization of Vegetable Production in Hydroculture Environments Using Artificial Intelligence: A Literature Review

Dick Díaz-Delgado | Ciro Rodríguez | Augusto Bernuy-Alva | Carlos Navarro | Alexander Inga-Alva

Environmental Science, Computer Science · Sustainability · 31 March 2025

TLDR AI-driven optimization in hydroculture improves nutrient **management**, water efficiency, and plant health monitoring, leading to higher yields and sustainability, despite its benefits, challenges such as data availability, model standardization, and implementation costs persist. Expand

9 Publisher Save Cite

Adaptive Hierarchical Multi-Headed Convolutional Neural Network With Modified Convolutional Block Attention for Aerial Forest Fire Detection

Md. Najmul Mowla | D. Asadi | S. Masum | Khaled M. Rabie

Environmental Science, Computer Science · IEEE Access · 2025

TLDR This study introduces a novel architecture, namely, the Adaptive Hierarchical Multi-Headed Convolutional Neural Network with Modified Convolutional Block Attention Module (AHMHCNN-mCBAM), which addresses prior challenges by integrating adaptive pooling, concatenated convolutions for multi-scale feature extraction, and an improved attention mechanism incorporating shared fully connected layers. Expand

11 PDF IEEE Save Cite

Detecting Cybersecurity Attacks in Internet of Things Using Artificial Intelligence Methods: A Systematic Literature Review

Mujaheed Abdullahi | Yahia Baashar | +4 authors | S. J. Abdulkadir

Computer Science, Engineering · Electronics · 10 January 2022

TLDR A systematic literature **review** that categorize, map and **survey** the existing literature on AI methods used to detect cybersecurity attacks in the IoT environment and provides an insight into the AI roadmap to detect threats based on attack categories is presented. Expand

270 [PDF] PDF Save Cite

A Survey of COVID-19 Diagnosis Using Routine Blood Tests with the Aid of Artificial Intelligence Techniques

Soheila Abbasi Habashi | M. Koyuncu | R. Alizadehsani | Medicine, Computer Science · Diagnostics · 1 May 2023

TLDR This study **review**ed some newly emerging **artificial intelligence** (AI)-based methods to diagnose COVID-19 using routine blood tests and concluded that Random **Forest** and logistic regression are the most widely used **machine learning** methods and the most commonly used performance metrics are accuracy, sensitivity, specificity, and AUC. Expand

7 [PDF] PDF Save Cite

Exploring reinforcement learning in process control: a comprehensive survey

N. Rajasekhar | T. K. Radhakrishnan | N. Samsudeen | Engineering, Computer Science · International Journal of Systems Science · 2 March 2025

TLDR This **review** paper presents a comprehensive examination of the development, models, algorithms and practical uses of RL, with a specific emphasis on its application in process control. Expand

30 Taylor & Francis Save Cite

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